

0. Assignment Overview

The assignment examines trip generation using three complementary approaches:

1. Develop and compare household-based and zone-based linear regression models for daily work trips.
2. Develop a household-based linear regression model for daily non-work trips and compare it with the household work-trip model.
3. Develop a cross-classification model for daily non-work trips using household size and vehicle ownership.

This first stage is limited to project setup, data loading, and basic validation. Modeling and exploratory analysis will be developed in later stages.

1. Setup and Imports

Project Paths

The project root is identified from the current working directory so that the notebook can run from either the repository root or the `notebooks` directory.

2. Data Loading

Each row represents one surveyed household. The expansion factor (`expfac`) is retained as provided but is not applied during this initial inspection.

3. Data Inspection and Validation

The checks below confirm the dataset dimensions, variable names, data types, memory use, and missing-value counts.

```
pd.DataFrame({"column": households.columns})
```

	column
0	zone
1	dwtype
2	npers
3	nveh
4	nlic
5	nftw
6	nptw
7	nwah
8	nstud
9	nfem
10	nmale
11	nchild
12	n65+
13	nwork
14	nnwk
15	expfac

```
households.dtypes.rename("dtype").to_frame()
```

	dtype
zone	int64
dwtype	int64
npers	int64
nveh	int64
nlic	int64
nftw	int64
nptw	int64
nwah	int64
nstud	int64
nfem	int64
nmale	int64
nchild	int64
n65+	int64
nwork	int64
nnwk	int64
expfac	float64

```
households.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2310 entries, 0 to 2309
Data columns (total 16 columns):
#   Column      Non-Null Count  Dtype
---  -
0    zone        2310 non-null   int64
1    dwtype      2310 non-null   int64
2    npers       2310 non-null   int64
3    nveh        2310 non-null   int64
4    nlic        2310 non-null   int64
5    nftw        2310 non-null   int64
6    nptw        2310 non-null   int64
7    nwah        2310 non-null   int64
8    nstud       2310 non-null   int64
9    nfem        2310 non-null   int64
10   nmale       2310 non-null   int64
11   nchild      2310 non-null   int64
12   n65+        2310 non-null   int64
13   nwork       2310 non-null   int64
14   nnwk        2310 non-null   int64
15   expfac      2310 non-null   float64
dtypes: float64(1), int64(15)
memory usage: 288.9 KB
```

```
missing_values = (
    households.isna()
    .sum()
    .rename("missing_values")
    .to_frame()
)
missing_values
```

	missing_values
zone	0
dwtype	0
npers	0
nveh	0
nlic	0
nftw	0
nptw	0
nwah	0
nstud	0
nfem	0
nmale	0
nchild	0
n65+	0
nwork	0
nnwk	0
expfac	0

```
expected_columns = [  
    "zone", "dwtype", "npers", "nveh", "nlic", "nftw", "nptw",  
    "nwah", "nstud", "nfem", "nmale", "nchild", "n65+", "nwork",  
    "nnwk", "expfac",  
]  
  
assert not households.empty, "The dataset is empty."  
assert households.columns.tolist() == expected_columns, "Unexpected dataset columns."  
  
print("Validation passed: the dataset is non-empty and contains all 16 expected columns.")
```

Validation passed: the dataset is non-empty and contains all 16 expected columns.